

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - MEDIUM (Assignment)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 3

Total Marks: 30

ANNEXURE A

Assignment

Code of Conduct:

I Kandath Haneesh (192324084) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Kandath Haneesh

ANNEXURE

Assignment

1. Online Shopping Coupon Verification

"An e-commerce website offers the coupon code SAVE50. Write a Java program that:

Removes extra spaces using trim()

Converts the entered code to uppercase using toUpperCase()

Compares it with the original coupon using equalsIgnoreCase()

The screenshot displays a web-based coding environment. On the left, a sidebar titled 'QUESTIONS' lists three items: 'Q1 Inventory Product Search System' (10 marks), 'Q2 Online Shopping Coupon Verification' (10 marks), and 'Q3 University Course Enrollment Manager' (10 marks). The 'Q2' question is selected and expanded. The main area shows the question text: 'An e-commerce website offers the coupon code SAVE50. Write a Java program that: Removes extra spaces using trim(), Converts the entered code to uppercase using toUpperCase(), Compares it with the original coupon using equalsIgnoreCase()'. Below the question, there is a 'Your Code' section with a text editor containing a Java program. To the right of the code editor is an 'Input' field with the value 'save50' and an 'Output' section showing the program's execution results. At the bottom of the code editor are 'Run' and 'Save' buttons.

QUESTIONS

- Q1** Inventory Product Search System 10 marks
- Q2** Online Shopping Coupon Verification 10 marks
- Q3** University Course Enrollment Manager 10 marks

3/3 answered

Q2 Online Shopping Coupon Verification 10 marks

An e-commerce website offers the coupon code SAVE50. Write a Java program that:

- Removes extra spaces using trim()
- Converts the entered code to uppercase using toUpperCase()
- Compares it with the original coupon using equalsIgnoreCase()

Your Code

```
1 import java.util.Scanner;
2 public class CouponVerification {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String coupon = "SAVE50";
6         System.out.print("Enter coupon code: ");
7         String code = sc.nextLine();
8         code = code.trim();
9         System.out.println("Uppercase Code: " + code.toUpperCase());
10        if (coupon.equalsIgnoreCase(code)) {
11            System.out.println("Coupon is Valid");
12        } else {
13            System.out.println("Invalid Coupon");
14        }
15        sc.close();
16    }
17 }
```

Input

save50

Output

```
Enter coupon code: Uppercase Code:
SAVE50
Coupon is Valid
```

Run **Save**

2. Inventory Product Search System

"A supermarket stores product names in different letter cases. Develop a program that:

Converts product names to lowercase using `toLowerCase()`

Searches a product using `contains()`

Confirms exact matching using `equalsIgnoreCase()`"

QUESTIONS

Q1
Inventory Product Search System
10 marks

Q2
Online Shopping Coupon Verification
10 marks

Q3
University Course Enrollment Manager
10 marks

3/3 answered

Q1 Inventory Product Search System10 marks

A supermarket stores product names in different letter cases. Develop a program that:

Converts product names to lowercase using `toLowerCase()`
Searches a product using `contains()`
Confirms exact matching using `equalsIgnoreCase()`

Your Code

```
1 import java.util.Scanner;
2 public class ProductSearch {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter product name: ");
6         String product = sc.nextLine();
7         String lower = product.toLowerCase();
8         System.out.println("Lowercase Product Name: " + lower);
9         System.out.print("Enter word to search: ");
10        String search = sc.nextLine();
11        if (lower.contains(search.toLowerCase())) {
12            System.out.println("Product contains the search word.");
13        } else {
14            System.out.println("Product does not contain the search word.");
15        }
16        System.out.print("Enter product name for exact match: ");
17        String check = sc.nextLine();
18        if (product.equalsIgnoreCase(check)) {
19            System.out.println("Exact Match Found.");
20        } else {
21            System.out.println("No Exact Match.");
22        }
23        sc.close();
24    }
25 }
```

Input

LaPtop
top
laptop

Output

Enter product name: Lowercase
Product Name: laptop
Enter word to search: Product
contains the search word.
Enter product name for exact
match: Exact Match Found.

3. University Course Enrollment Manager

"A university maintains a list of students enrolled in a course. Students who withdraw must be removed from the list.

Requirements:

Add student names to an ArrayList

Traverse the list using an Iterator

Remove withdrawn students and display the updated enrollment list"

QUESTIONS

Q1
Inventory Product Search System
10 marks

Q2
Online Shopping Coupon Verification
10 marks

Q3
University Course Enrollment Manager
10 marks

3/3 answered

Q3 University Course Enrollment Manager10 marks

A university maintains a list of students enrolled in a course. Students who withdraw must be removed from the list.

Requirements:

Add student names to an ArrayList

Traverse the list using an Iterator

Remove withdrawn students and display the updated enrollment list

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.Scanner;
4 public class CourseEnrollment {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         ArrayList<String> students = new ArrayList<>();
8         System.out.print("Enter number of students: ");
9         int n = sc.nextInt();
10        sc.nextLine();
11        for (int i = 0; i < n; i++) {
12            System.out.print("Enter student name: ");
13            students.add(sc.nextLine());
14        }
15        System.out.print("Enter withdrawn student name: ");
16        String withdrawn = sc.nextLine();
17        ArrayList<String> updated = new ArrayList<>();
18        Iterator<String> it = students.iterator();
19        while (it.hasNext()) {
20            String name = it.next();
21            if (!name.equalsIgnoreCase(withdrawn)) {
22                updated.add(name);
23            }
24        }
25        System.out.println("Updated Enrollment List:");
26        for (String name : updated) {
27            System.out.println(name);
28        }
29        sc.close();
30    }
31 }
```

Input

```
Mehi
Aftab
Prakash
Ravi
```

Output

```
student name: Mehi withdrawn
student name: Updated Enrollment List:
Mehi
Aftab
Prakash
```

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification